

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - IX

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Class: TYIT	Batch: 1
Date of Assignment: 18-08-2026	Date/Time of Submission: 18-08-2026

a. Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

[Student.cs](#)

```
namespace practical9A.Models
```

```
{  
    public class Student  
    {  
        public int Id { get; set; }  
        public string Name { get; set; }  
        public int Age { get; set; }  
        public string Course { get; set; }  
    }  
}
```

[AppDbContext.cs](#)

```
using Microsoft.EntityFrameworkCore;
```

```
using practical9A.Models;
```

```
namespace practical9A.Data
```

```
{  
    public class AppDbContext:DbContext  
    {  
        public AppDbContext(DbContextOptions<AppDbContext> options):base(options)  
        {  
        }  
        public DbSet<Student> Student { get; set; }  
    }  
}
```

[Program.cs](#)

```
using Microsoft.EntityFrameworkCore;
```

```
using StudentAPI.Data;
```

```

var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllers();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI();

app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();

app.Run();

```

```

StudnetController.cs
using Microsoft.AspNetCore.Mvc;
using StudentAPI.Data;
using StudentAPI.Models;

namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly AppDbContext _context;

        public StudentsController(AppDbContext context)
        {
            _context = context;
        }

        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }
    }
}

```

```

}

// GET BY ID
[HttpGet("{id}")]
public IActionResult GetStudent(int id)
{
    var student = _context.Students.Find(id);

    if (student == null)
        return NotFound();

    return Ok(student);
}

// INSERT
[HttpPost]
public IActionResult AddStudent(Student student)
{
    _context.Students.Add(student);
    _context.SaveChanges();

    return Ok(student);
}

// UPDATE
[HttpPut("{id}")]
public IActionResult UpdateStudent(int id, Student student)
{
    var data = _context.Students.Find(id);

    if (data == null)
        return NotFound();

    data.Name = student.Name;
    data.Age = student.Age;
    data.Course = student.Course;

    _context.SaveChanges();

    return Ok(data);
}

// DELETE
[HttpDelete("{id}")]

```

```

public IActionResult DeleteStudent(int id)
{
    var student = _context.Students.Find(id);

    if (student == null)
        return NotFound();

    _context.Students.Remove(student);
    _context.SaveChanges();

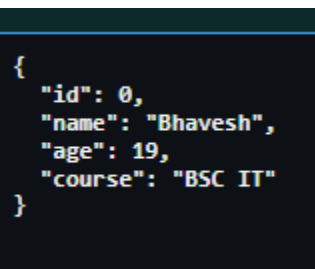
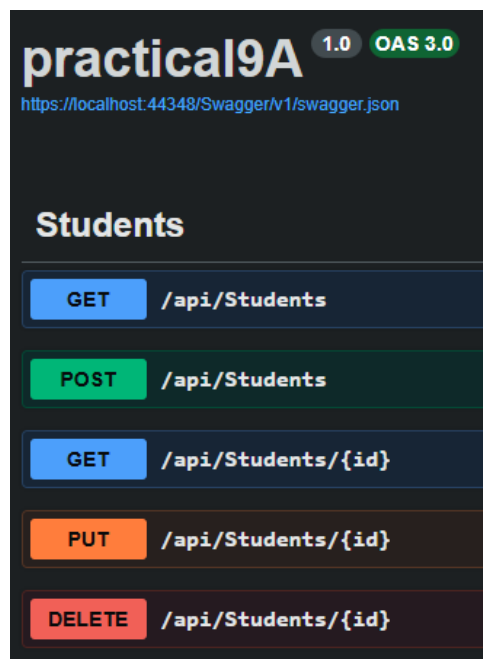
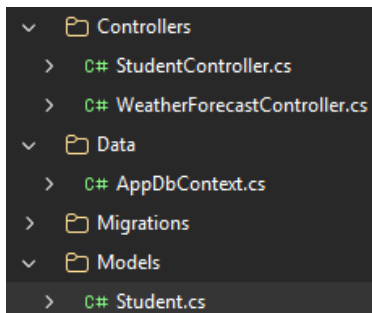
    return Ok();
}
}
}

```

```

PM> Add-Migration InitialCreate
Build started...
Build succeeded.
To undo this action, use Remove-Migration.
PM> Update-Database
Build started...
Build succeeded.

```



Code

Details

200

Response body

```

{
  "id": 4,
  "name": "Bhavesh",
  "age": 19,
  "course": "Bsc IT"
}

```

Response headers

```

content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 03:30:10 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET

```

Responses

Code	Description
200	OK

Code

Details

200

Response body

```

{
  "id": 1,
  "name": "Bhavesh",
  "age": 19,
  "course": "BSC IT"
}

```

Response headers

```

content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 03:32:31 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET

```

Responses

Code	Description
200	OK

←

→

↺

🌐

localhost:44348/api/Students/1

Pretty-print ☒

```

{
  "id": 1,
  "name": "Bhavesh",
  "age": 19,
  "course": "BSC IT"
}

```

PUT /api/Students/{id}

Parameters

Name	Description
id * required	integer(\$int32) (path)

1

Request body

Edit Value | Schema

```

{
  "id": 6,
  "name": "Bhavesh",
  "age": 18,
  "course": "Bsc IT"
}

```

dbo.Students [Data] ✕ dbo.Students [Design] dbo.Students [Des]				
	Id	Name	Age	Course
	1	Bhavesh	18	Bsc IT
	2	Bhavesh	19	Bsc IT
	4	Bhavesh	19	Bsc IT
	5	Bhavesh	19	Bsc IT
	6	Bhavesh	19	Bsc IT
	NULL	NULL	NULL	NULL
	NULL	NULL	NULL	NULL

DELETE /api/Students/{id}

Parameters

Name	Description
id * required	integer(\$int32) (path)

3

	Id	Name	Age	Course
▶	1	Bhavesh	18	Bsc IT
	2	Bhavesh	19	Bsc IT
	3	Bhavesh	19	Bsc IT
	4	Bhavesh	19	Bsc IT
	5	Bhavesh	19	Bsc IT
	6	Bhavesh	19	Bsc IT
⊕	NULL	NULL	NULL	NULL

b. Create an API endpoint to calculate Factorial of a Number and test it using Postman.

Code:

```
using Microsoft.AspNetCore.Mvc;
```

```
namespace FactorialAPI.Controllers
```

```
{
```

```
    [Route("api/[controller]")]
```

```
    [ApiController]
```

```
    public class MathController : ControllerBase
```

```
    {
```

```
        [HttpGet("factorial/{number}")]
```

```
        public IActionResult GetFactorial(int number)
```

```
        {
```

```
            if (number < 0)
```

```
            {
```

```
                return BadRequest("Factorial not possible for negative numbers.");
```

```
            }
```

```
            long factorial = 1;
```

```
            for (int i = 1; i <= number; i++)
```

```
            {
```

```
                factorial *= i;
```

```
            }
```

```
            return Ok(new
```

```
            {
```

```
                Number = number,
```

```
                Factorial = factorial
```

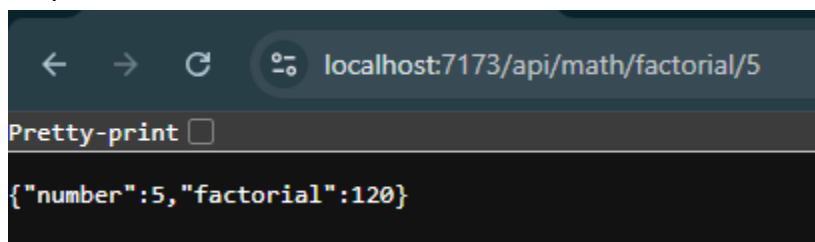
```
            });
```

```
        }
```

```
    }
```

```
}
```

Output:





https://localhost:7173/api/math/factorial/5

GET



https://localhost:7173/api/math/factorial/5

1
2
3
4



```
"number": 5,  
"factorial": 120
```